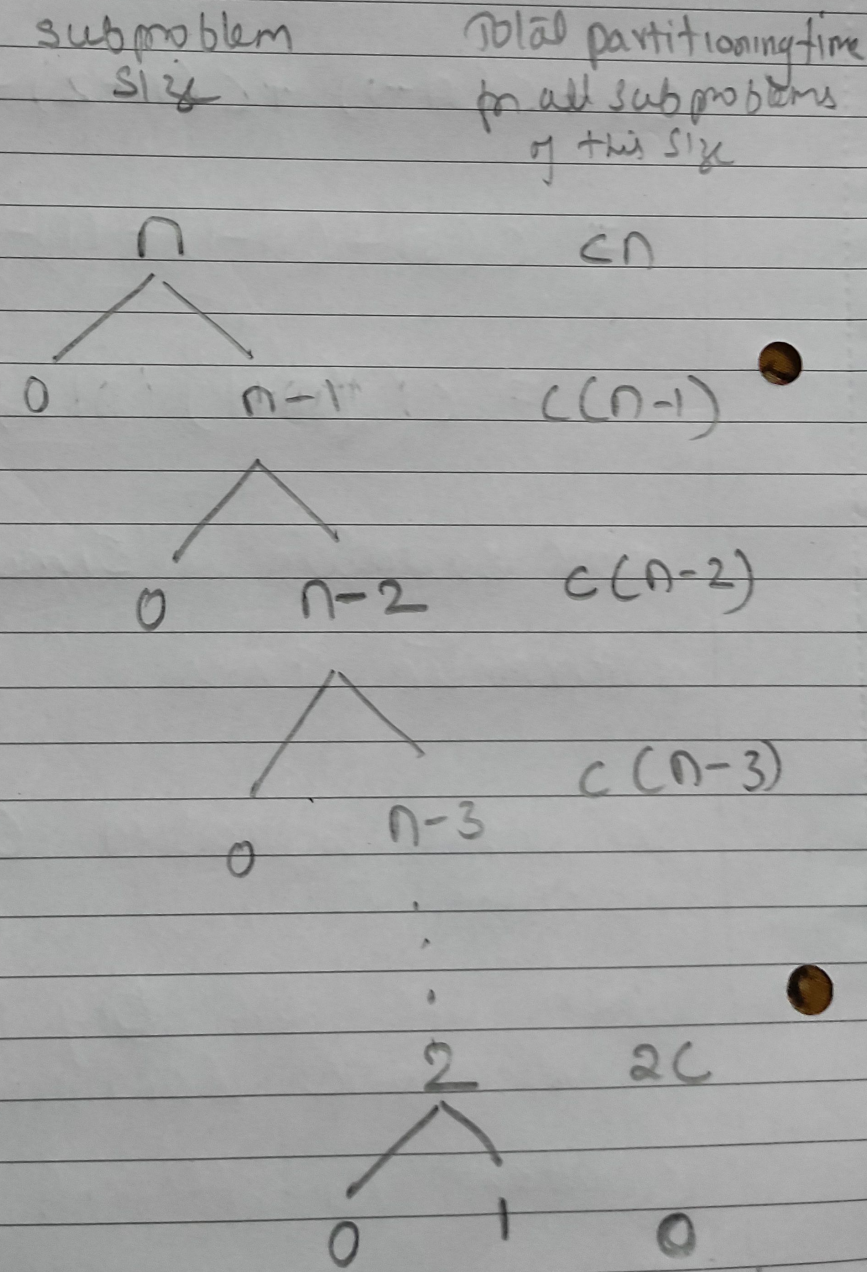
Worst case

When quicksort always has the most unbalanced partitions possible then original call takes cn time for some constant c, the recursive call on $(n-1)$ elements takes $c(n-1)$ time, the recursive call on $n-2$ elements $c(n-2)$ time, and so on.

Here's a tree of the subproblem sizes with their partitioning times:



When we total up the partitioning times for each level, we get

$$cn + c(n-1) + c(n-2) + \cdots + 2c + c$$

$$= c(n + (n-1) + (n-2) + \cdots + 2 + 1)$$

$$= c\left(\frac{(n+1)(n+2)}{2} - 1\right)$$

Our calculation is similar to arithmetic series.

In above expression C is constant co-efficients

When we use big O notation, we ignore low order terms and constant co-efficients.

So in big O notation our worst case running time is

$O(n^2)$